accelerator design at CERN—was of the very first rank.

Bell also left behind a research program in quantum foundations. The year before he died, at a conference in the mountain village of Erice on the western tip of Sicily, Bell "delivered what came close to being the most spell-binding lecture I have ever heard," Mermin later recalled. "What exactly qualifies some physical systems to play the role of 'measurer'?" Bell asked, dripping with sarcasm. "Was the wave function of the world waiting to jump for thousands of millions of years until a single-celled living creature appeared? Or did it have to wait a little longer, for some better qualified system… with a PhD?" Bell then pointed out the flaws in how quantum physics was commonly taught (taking several specific textbooks to task for their lapses in explanation) and finally laid out the two approaches to quantum physics that he thought were the most promising: the pilot-wave interpretation and spontaneous-collapse theory. He concluded his talk with a challenge. "The big question, in my opinion, is which, if either, of these two precise pictures can be redeveloped in a [way consistent with special relativity]."

But Bell also entertained a third option, beyond pilot waves and spontaneous collapse, though he did not mention it in his talk at Erice. "The 'many world interpretation' seems to me an extravagant, and above all an extravagantly vague, hypothesis," Bell said in 1986. "I could almost dismiss it as silly. And yet.… It may have something distinctive to say in connection with the 'Einstein Podolsky Rosen puzzle,' and it would be worthwhile, I think, to formulate some precise version of it to see if this is really so. And the existence of all possible worlds may make us more comfortable about the existence of our own world… which seems to be in some ways a highly improbable one." Though the number of physicists interested in pilot waves and spontaneous collapse grew after Bell's death, the many-worlds interpretation gained far more popularity and notoriety in the last decades of the twentieth century. And much of the reason had nothing to do with Bell's work, or any work in quantum physics at all. Instead, many-worlds came roaring back thanks to research in an entirely different field of physics, a field that studied not the ridiculously small but the unthinkably huge: the entire universe itself.